Numerical Optimization for VITESS simulations- Version 2

Principle

The combination of Monte Carlo simulations and optimization is realized in a way that the optimization routine is the main program and calls the simulation in a subroutine called *ExtFunctions()* (see Fig. 1).
This procedure first writes a file *Pcomm.dat* containing one or more parameter sets. The tool *gener_pipe.exe* reads these sets and combines them with the standard instrument (saved as *std_instr.cmd*) , thus generating the VITESS pipe commands (in a shell or batch script) for all simulations to perform.
The simulations are then started and the files (of all simulations) that are needed for the figure of merit are copied. The next step is to start a program to calculate the figures-of-merit (for all simulations) and write them into a file *Fcomm.dat*. Now this/these figure(s)-of-merit are read and returned to the optimization program. This will suggest (a) new parameter set(s) and call *ExtFunctions* again. This procedure continues until the optimization has come to an end.
The optimization is widely controlled by the 3 ASCII input files '*opt_param.ini*', '*sim_param.ini*' and '*fom.ini*' and an ASCII control file for the optimization used. If there is a 1 to 1 relation between optimization and simulation parameters and the figure of merit is one of the many options of the delivered *fom* program, no code or script is necessary for the optimization.

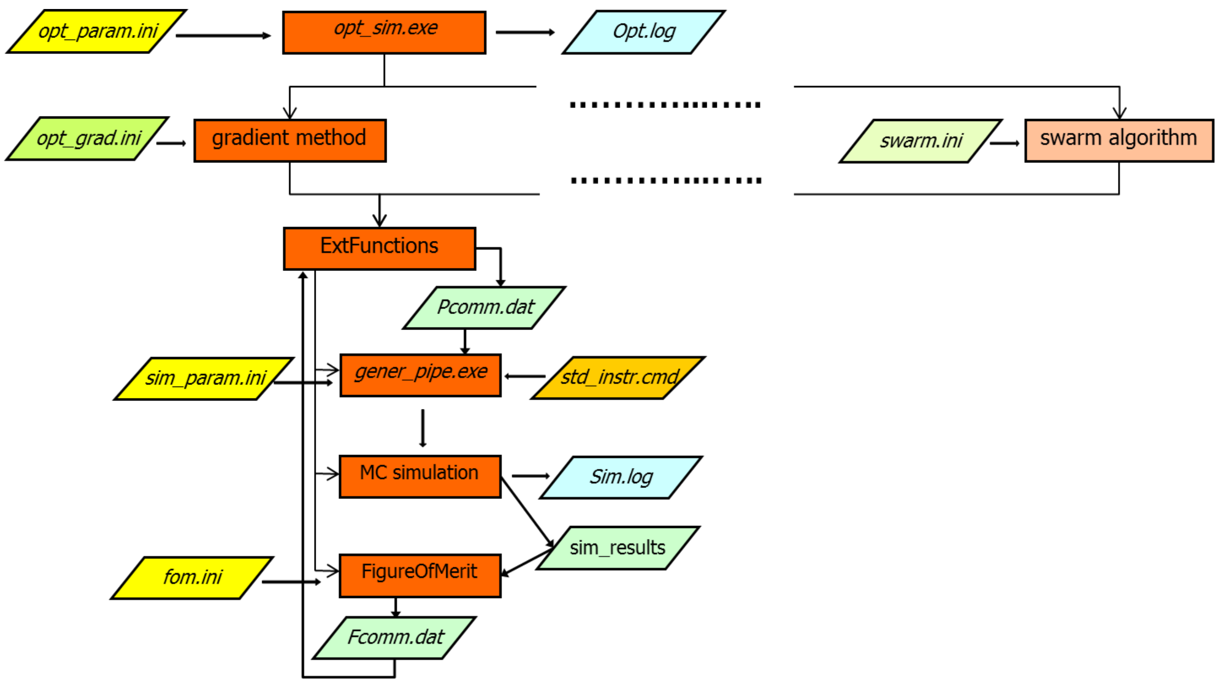

Figure 1: Scheme of the optimization

Directories

The directory '… VitessX.Y/OPTIMIZATON' contains all necessary executables and input files for the optimization. You can run the optimization here or copy all files to another directory and run it there. This will be called '**optimization directory**'.
The simulations generally run in a different directory. This can be the directory where the instrument has been developed, or – if you choose another directory (cf. step 3) – you have to copy all input files there. This will be called '**simulation directory**'.

6 Steps to the Optimization

Step 1: Standard Instrument

The first step is to develop an instrument that can be used as a starting point for the optimization. Make sure that the simulation runs properly and that an adequate number of trajectories is chosen; 500000 – 1000000 trajectories should contribute to the figure of merit, i.e. have to be summed up in the file used to calculate the figure of merit.

When it is finished it needs to be saved by clicking "Check" and saving the shown pipe command as *std_instr.cmd*. to the **optimization directory** !!! Note that this command cannot be used for an optimization on another platform, (e.g. saved under Windows and used under Unix).

Now the different ini-files have to be filled. Note that anything behind a #-symbol is regarded as a comment and ignored by the program.

Step 2: Optimization Parameters

In the file '*opt_param.ini*' you have to give

1. the name of the optimization
2. a code word to define the optimization algorithm
   existing at the moment : 'opt_grad', 'opt_grad_mc', 'metropolis'
3. option for parallel computing
4. all input variables in the form
   - index (starting with 1)
   - initial value
   - minimal value
   - maximal value
   - difference to the actual value for the calculation of the gradient or the next tested value

Notes:

- Not all algorithms will need all information about the parametrs or can treat them, but for the sake of simplicity this needs to be given for all of them.
- Input lines must not be removed and their order has to be kept.

```
Optimisation of the guide exit for EXED     # name of the optimization
#
opt_grad_mc    #  algorithm,         existing: 'opt_grad', 'opt_grad_mc', 'metropolis'
parallel       #  parallel. option, existing: 'parallel', 'sequential'
#
# parameters
#  start    min      max   delta     description
1    2.0   -10.0    10.0   0.05    # hor. shift             [cm]
2    2.0   -10.0    10.0   0.05    # vert. shift            [cm]
```

File 1: file *opt_param.ini* setting optimization algorithm and parameters to be optimized

Step 3: Control of the Simulation

The first 2 lines in *sim_param.ini* define the directory where the executables can be found (and thus the Vitess version) – line 1 – and the simulation directory (cf. section 'Directories') – line 2 (see Files 2a and 2b). STD means that the path to this directory is taken from the standard instrument file 'std_instr.cmd'
The third line defines the file(s) necessary to calculate the figure of merit are (see File 2a).

The next line of this file defines the function calculating the simulation parameters S_k from the optimization parameters P_j. STD defines a direct assignment: S0 = P0, S1=P1, etc. In the future, simple mathematical operations between these parameters can be defined in the way indicated in the file. For the time being, only this direct assignment is possible, everything else has to be realized by extending the program *gener_pipe.exe* - see below.

In the next lines, all simulation parameters affected by the optimization parameters are listed (see File 2a,b). They are characterized by the module number and the symbol (like "-H") defining the parameter (within the list of existing parameters of this module). The symbol is shown if you click on the text belonging to this item.

```
# -----------------------------------------------------------------------------------------
STD                  # directory of VITESS executables       STD: directory from 'std_instr.cmd
STD                  # parameter directory of the simulation STD: directory from 'std_instr.cmd
# -----------------------------------------------------------------------------------------
sample.mtl           # files to be copied for 'FigureOfMerit'
# -----------------------------------------------------------------------------------------
STD                  # defines function to transfer optimization parameters to simulation parameters
# SimPar  FitPar       ------------------------------------------------------------------------
  6 -y   = P(1)      # hor. shift          [cm]
  6 -z   = P(2)      # vert. shift         [cm]
# -----------------------------------------------------------------------------------------
```

File 2a: file *sim_param.ini* defining the variable simulation parameters and the function transferring the optimization parameters to simulation parameters.

In order to calculate simulation parameters from optimization parameters, a function similar to *DetSimParamStd*() has to be written in '*gener_pipe.c*, which performs this calculation. (The easiest and safest way is the following: copy, rename and change *DetSimParamStd*().

Then add 3 lines of code in the *if…else if…else* command identifying this function by code that will then be given in *sim_param.ini*.instead of STD.
Example: Adding

```
else if (strcmp(sParFct, "MTR02")==0)
{       DetSimParamMtr(P, m, nSimParNo, nFitParNo,  2);
}
```

assigns that the function *DetSimParamMtr()* with additional parameter 2 is called to transfer the optimization parameters to simulation parameters, if the code 'MTR02' is given in the forth line of *sim_param.ini*. (This has in fact been realized for the optimization of the elliptical guide exit of the EXED instrument, cf. File 2b. MTR denotes the change from meter to cm and 02 indicates to which simulation parameters this is applied.)

```
sim_param.ini - Editor
Datei  Bearbeiten  Format  Ansicht  ?
# -------------------------------------------------------------------------------------------------------
STD                                # directory of VITESS executables       STD: directory from 'std_instr.cmd
N:/VitessOpt/FitSimulation/Test2   # parameter directory of the simulation STD: directory from 'std_instr.cmd
# -------------------------------------------------------------------------------------------------------
sample.mtl                         # files to be copied for 'FigureOfMerit'
MTR02                              # defines function to transfer optimization parameters to simulation parameters
# -------------------------------------------------------------------------------------------------------
# SimPar  FitPar
   3 -W = X(1)      # exit width           [cm]
   3 -H = X(2)      # exit height          [cm]
   3 -f = X(3)*100 # hor. focusing point   [cm]
   3 -F = X(4)*100 # vert. focusing point  [cm]
```

File 2b: special version of *sim_param.ini* where the function DetSimParamMtr is used for the transfer.

Step 4: The Figure of Merit

The next step is to assess the figure of merit. The function '*ExtFuncitons()*' will call the batch/shell script *FigureOfMerit.bat* and expects the results in the file *Fcomm.dat*. It is an ASCII file containing just one number for each simulation to be executed. In case of N simulations, these N numbers are expected in N rows. The files defined in *sim_param.ini* – in this example *sample.mtl* - serve as a data basis for these calculations. In principle any kind of program can be used to calculate the values from these input file(s) and write them into *Fcomm.dat*. You just has to be called it in *FigureOfMerit.bat*.

To avoid repetition of work a routine called *fom.exe* is delivered with the package. By default this is called by *FigureOfMerit.bat*. It is controlled via *fom.ini*. The general formulae are

$$FoM = \frac{\sum_i \frac{w_i I_{sign,i} \lambda_i^l}{I_{ref,i}^m}}{\sum_i I_{noise,i}^n}$$

or

$$FoM = \frac{< \frac{w_i I_{sign,i} \lambda_i^l}{I_{ref,i}^m} >}{< I_{noise,i}^n >}$$

Where the parameter A defines if sums (A=0) or averages (A=1) are used. The numbers for the powers l,m,n have to be given; they can be float values. The files containing signal, noise,

reference and weight have to be given. (If they output files of the simulations they have to be listed in *sim.ini* as well. If a weighting with wavelength shall be used, the intensity files have to be intensities as a function of wavelength.
As the optimization routine searches for the minimum, the value f/FoM is written to *Pcomm.dat*. The factor is also read from fom.ini and allows setting output values to the order of magnitude 1, which is an advantage (in terms of numerical precision) for some optimization algorithms.
In the example shown in File 7, the figure of merit is reduced to

$$FoM = \sum I_{sign,i}$$

thus optimizing the integrated flux (in the slit).
Another example: if the transmission averaged over a wavelength range shall be optimized, the parameters have to be set as: A=1; m=1; l=n=0 and the reference file (=input of the section, whose transmission is regarded) has to be given in addition to the signal file.

```
fom.ini - Editor
Datei  Bearbeiten  Format  Ansicht  ?
A= 0          # criterion: sum (A=0) or average (A=1)
l= 0.0        #        Sum\Average(I_sig/I_ref^m * Lambda^l)
m= 0.0        #  FoM = ------------------------------------
n= 0.0        #            Sum\Average(I_noise)^n
f= 1.0e09     # normalization factor for output value: f/FoM
S=slit.mtl    # signal file for 'FigureOfMerit'
R=no_file     # reference file
N=no_file     # noise file
W=no_file     # weight file
```

File 3: file *fom.ini* to define the figure of merit using the program *fom_lambda.exe*.

Step 5: Control of the optimization routine

The optimization process is controlled by the file 'opt_name.ini', i.e. each algorithm has a control file of its own. Useful default parameters are delivered so that usually no changes are necessary. Changing these parameters is possible, but requires knowledge of the algorithm.

```
opt_grad_mc.ini - Editor
Datei  Bearbeiten  Format  Ansicht  ?
# control parameters for 'opt_grad_mc'
# ------------------------------------
a= 2       # eOut   :  output control (1: only final result, 2: parameters in each step, 3: function values)
s=25       # mSteps:  max. number of optimization steps
n=10       # mDamps:  max. number of dampings
D=0.5      # TD     :  initial factor t in opt. step size betw. lin. and real fct.
d=0.5      # Damp   :  damping factor (reduction of t )
t=0.333    # Tfac   :  fraction Q-reduction in original and linearized fct.
r=1.000    # Qverm  :  ratio of Q-reduction within 1 step that stops the optimization
m=1.0E-24  # Qmin   :  Q-value to stop the optimization
R=0.8      # RDelpP:  reductionf factor of DelP in case of deterior. in both dir.
```

File 4: file *opt_grad_mc.ini* controlling a gradient method adapted to Monte Carlo simulations

Step 6: Run

Now the optimization can be started by calling the program *opt_sim* in the directory '… VitessX.Y/OPTIMIZATON'